Class: TE **Subject: AISC** **Sem-V**

Experiment No: 10

Course Outcome: Analyze fuzzy membership functions and IF–THEN rules to determine the appropriate washing duration based on dirt and grease levels.

Blooms Level: L4 – Analyze

Aim: To analyze the effect of dirt and grease levels on the washing duration using a fuzzy logic-based washing machine controller.

Abstract:

A washing machine must adjust its washing duration according to the type and intensity of stains present on clothes. Dirt and grease are two important factors that influence the cleaning process. Dirt may include dust, mud, and other particulate matter, whereas grease consists of oily or fatty substances that may require a longer washing cycle.

In this experiment, a fuzzy logic-based washing machine controller is developed using dirt level and grease level as input variables. Each input is classified into linguistic categories such as Low, Medium, and High using fuzzy membership functions. A set of IF–THEN rules determines the appropriate washing duration based on different combinations of dirt and grease levels. The fuzzy inference process combines the activated rules, and defuzzification converts the result into a numerical washing duration. The simulation enables students to analyze how variations in dirt and grease levels influence the washing cycle.

Sample Input and Output:

Case 1: Dirt level = 15, Grease level = 10 → Short washing duration

Case 2: Dirt level = 50, Grease level = 35 → Medium washing duration

Case 3: Dirt level = 85, Grease level = 90 → Long washing duration

Note: These are illustrative expected categories. Exact durations depend on the membership functions, rule base, and defuzzification method.

Theory:

Fuzzy logic is a computational approach that represents vague concepts using membership values between 0 and 1. In a washing machine, the amount of dirt and grease cannot always be classified using rigid thresholds. Fuzzy sets allow the controller to represent levels such as Low, Medium, and High and make a gradual decision about washing duration.

Input variables: Dirt level (0–100) and grease level (0–100). Both inputs are represented by fuzzy sets Low, Medium, and High. The output is washing duration from 10 to 60 minutes, represented by Short, Medium, and Long.

Sample rule base:

1. IF dirt is Low AND grease is Low, THEN washing duration is Short.
2. IF dirt is Medium AND grease is Low, THEN washing duration is Medium.
3. IF dirt is Low AND grease is High, THEN washing duration is Long.
4. IF dirt is High AND grease is Medium, THEN washing duration is Long.
5. IF dirt is Medium AND grease is High, THEN washing duration is Long.
6. IF dirt is High AND grease is High, THEN washing duration is Long.

The controller evaluates the rules using fuzzy inference and aggregates the output sets. Defuzzification then converts the combined fuzzy output into a single numerical duration. By testing different dirt and grease combinations, students can analyze the influence of each input on the washing cycle. This is a simplified educational simulation and not a calibrated model of a commercial washing machine.

Program (Python):

https://colab.research.google.com/drive/1w2Oc5njHn5cdrLV7QZw7C4KGqq-hwZrO?usp=sharing

Output:

Enter dirt percentage (0-100): 45

Enter grease percentage (0-100): 65

\nDirt membership values:

SD - Small Dirt: 0.10

MD - Medium Dirt: 0.90

\nGrease membership values:

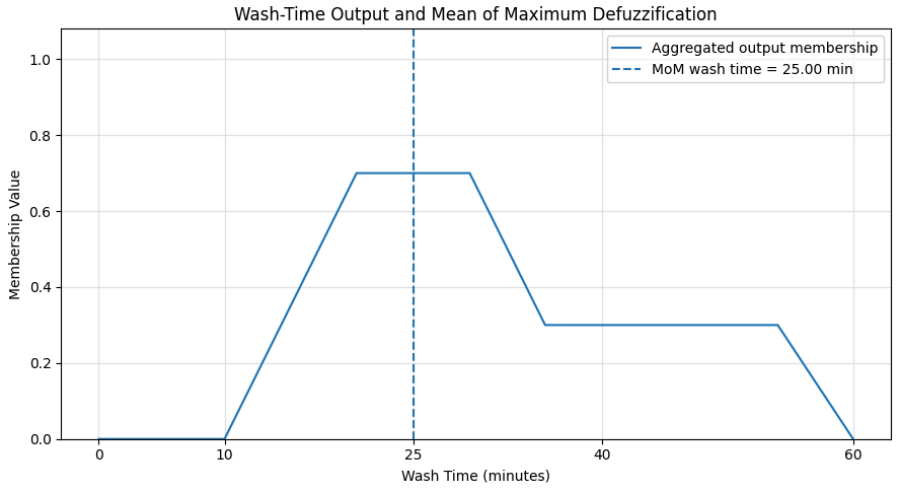
MG - Medium Grease: 0.70

LG - Large Grease: 0.30

\nActivated IF-THEN rules:

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IF SD - Small Dirt AND MG - Medium Grease THEN M - Medium; strength = 0.10
IF SD - Small Dirt AND LG - Large Grease THEN L - Large; strength = 0.10
IF MD - Medium Dirt AND MG - Medium Grease THEN M - Medium; strength = 0.70
IF MD - Medium Dirt AND LG - Large Grease THEN L - Large; strength = 0.30
\nRecommended wash time (MoM): 25.00 minutes



Conclusion:

The fuzzy logic controller models washing duration based on dirt and grease levels. By comparing different input combinations, students can analyze how the intensity of dirt and grease affects the output and evaluate the suitability of the fuzzy rules for different washing conditions.

Exercise 1:

Abstract:

Modify the controller to include fabric type as a third input. Define suitable fuzzy sets and update the rule base to recommend washing duration for delicate, normal, and heavy fabrics.

Exercise 2:

Abstract:

Extend the fuzzy controller to recommend both washing duration and water temperature using dirt and grease levels. Analyze the results for low, medium, and high input combinations.

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